

University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering

Problem Set 4.1: Specifications in the S-Plane

Question 1

Answer the following questions about the performance of the second-order systems with the given poles.

- i) How long will it take for the system to settle to within 5% of its final value?
- ii) What is the damping condition of the system (i.e. over-, critically or underdamped?) and what is the damping ratio.
- iii) What is the damped frequency of the system? (does not apply to every damping condition)
 - a) $-2 \pm 3j$
 - b) -4 and -8
 - c) Repeated pole at -1
 - d) 0 and -1
 - e) $-2 \pm j$

Question 2

Sketch the region of the s-plane that the poles must fall into to satisfy the following specifications:

- a) Settling to within 2% of the final value within 5 seconds
- b) A damping ratio of more than 0.8
- c) A damping ratio of more than 0.5 and damped frequency below 10 rad/s

Question 3

A spring-mass-damper system is represented by the differential equation

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + f(t) \quad (1)$$

where $x(t)$ is the horizontal position of the mass and $f(t)$ is an input force. For a damping coefficient of 8 and a spring constant of 12,

- a) select the lightest mass that will guarantee that the system settles to 5% of its final value within 2 seconds.
- b) Same as (a), but if the damping coefficient has a 10% tolerance.
- c) Work out the damping ratio of the system for your answer in (b).
- d) If the system is placed in a simple feedback loop (no controller, no sensor block), work out the sensitivity function for the system in terms of m , b and k . Which parameter affects the robustness of the closed-loop system the most? Does this change for low vs. high frequencies? (hint: you can input a frequency ω to a transfer function by substituting $s = j\omega$)

Question 4

- a) Some component's resistance has a sensitivity of 0.02 with respect to the temperature. If its resistance is 1 k Ω at 30°C, what will its resistance be if the temperature increases by 5°?
- b) Two quantities, x and y , are measured at two instances and found to have the following values:

Quantity	Measurement 1	Measurement 2
x	10	15
y	30	70

What is the sensitivity of y with respect to x ? Would you describe y as sensitive to x ?